CLASS ACTIVITY 20.07.26

1.

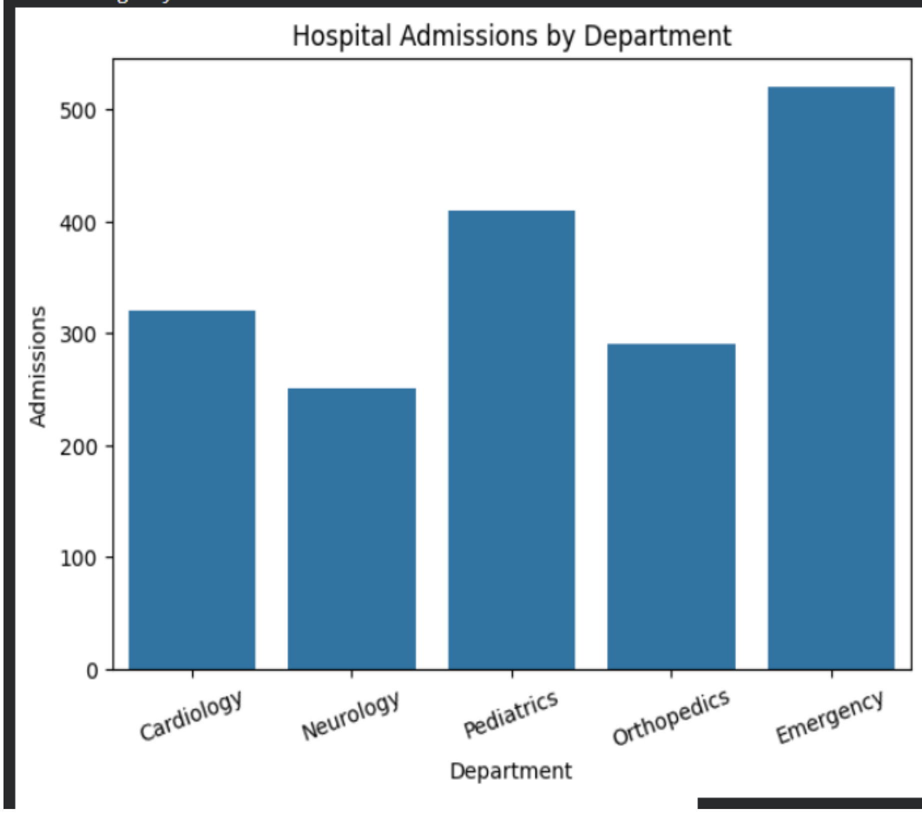
Python code :-

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
data = {
    "Department": [
        "Cardiology", "Neurology", "Pediatrics",
        "Orthopedics", "Emergency"
    ],
    "Admissions": [320, 250, 410, 290, 520]
}
df = pd.DataFrame(data)
print(df)
plt.figure(figsize=(7,5))
sns.barplot(x="Department", y="Admissions", data=df)
plt.title("Hospital Admissions by Department")
plt.xticks(rotation=20)
plt.show()
plt.figure(figsize=(5,5))
```

```
sns.boxplot(y=df["Admissions"])
plt.title("Admission Variability")
plt.show()
plt.figure(figsize=(6,4))
sns.histplot(df["Admissions"], bins=5, kde=True)
plt.title("Distribution of Admissions")
plt.show()
try:
    import squarify

    plt.figure(figsize=(7,5))
    squarify.plot(
        sizes=df["Admissions"],
        label=df["Department"],
        alpha=0.8
    )
    plt.title("Department Workload Treemap")
    plt.axis("off")
    plt.show()
except ImportError:
    print("Install squarify using:")
    print("pip install squarify")
```

Output:-

```
   Department  Admissions
0   Cardiology         320
1    Neurology         250
2   Pediatrics         410
3  Orthopedics         290
4    Emergency         520
```



R-code:-

library(ggplot2)

library(dplyr)

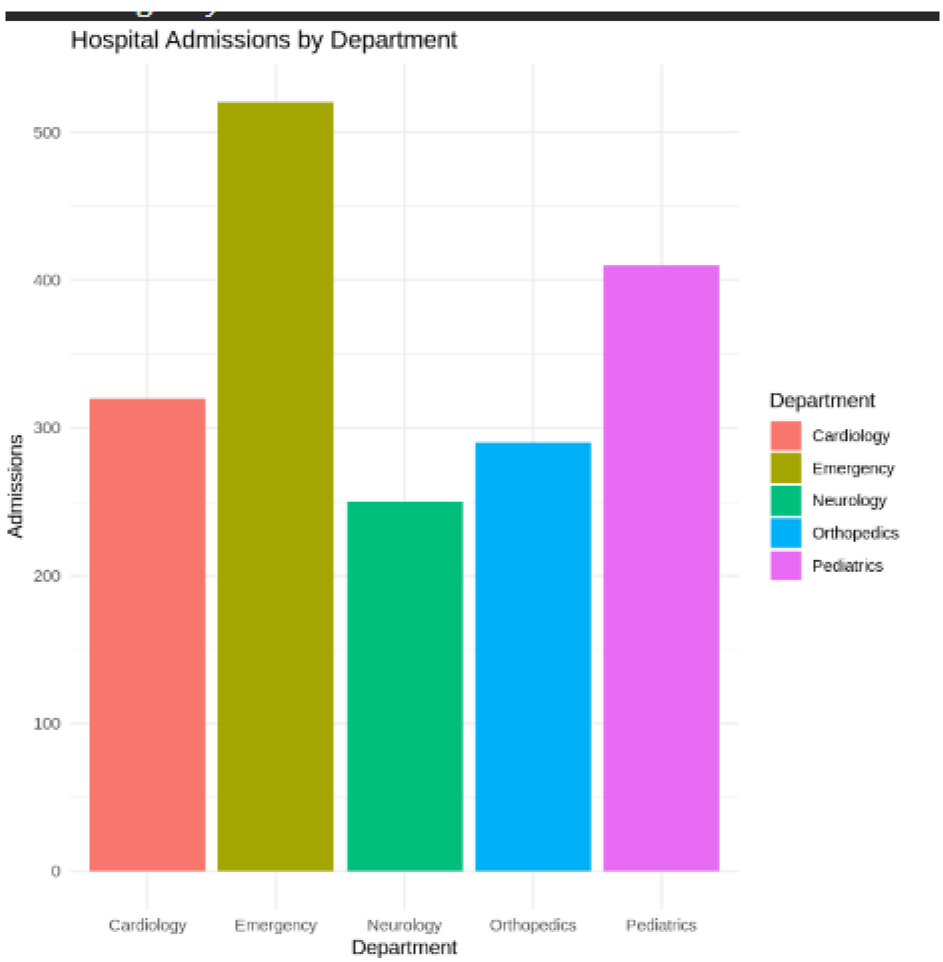
hospital <- data.frame(

  Department = c("Cardiology", "Neurology", "Pediatrics", "Orthopedics", "Emergency"),

```r
  Admissions = c(320, 250, 410, 290, 520)
)
print(hospital)
ggplot(hospital, aes(x = Department, y = Admissions, fill = Department)) +
  geom_bar(stat = "identity") +
  ggtitle("Hospital Admissions by Department") +
  theme_minimal()
ggplot(hospital, aes(y = Admissions)) +
  geom_boxplot(fill = "skyblue") +
  ggtitle("Admission Variability") +
  theme_minimal()
ggplot(hospital, aes(x = Admissions)) +
  geom_histogram(binwidth = 50, fill = "steelblue", color = "black") +
  ggtitle("Distribution of Admissions") +
  theme_minimal()
```

Output:-

	Department	Admissions
1	Cardiology	320
2	Neurology	250
3	Pediatrics	410
4	Orthopedics	290
5	Emergency	520



2.

Python code :-

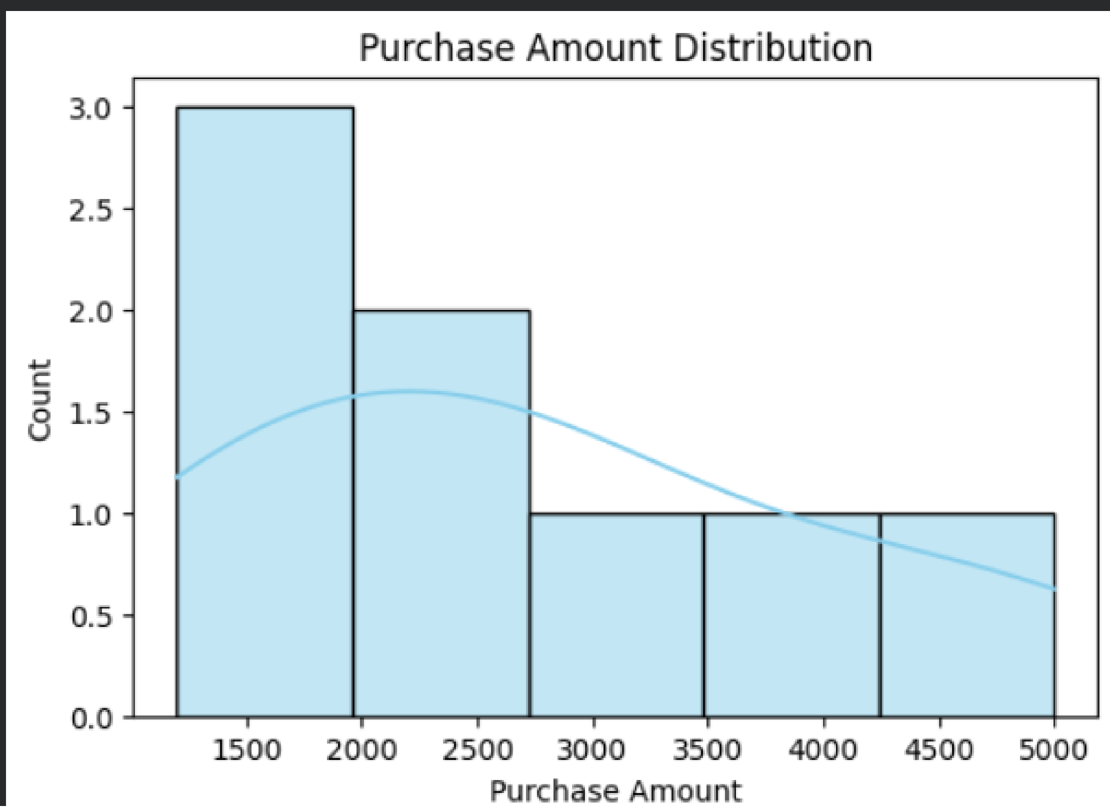
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
data = {
   "Customer": ["C1", "C2", "C3", "C4", "C5", "C6", "C7", "C8"],
   "PurchaseAmount": [1200, 2500, 1800, 4200, 3100, 1500,
5000, 2700],
   "OrderFrequency": [5, 8, 6, 12, 10, 4, 15, 7],
   "Category": ["Electronics", "Fashion", "Electronics",
"Home",
             "Fashion", "Home", "Electronics", "Fashion"]
}


df = pd.DataFrame(data)
print(df)
plt.figure(figsize=(6,4))
sns.histplot(df["PurchaseAmount"], bins=5, kde=True,
color="skyblue")
```

```
plt.title("Purchase Amount Distribution")
plt.xlabel("Purchase Amount")
plt.show()
plt.figure(figsize=(6,4))
sns.violinplot(y=df["PurchaseAmount"], color="lightgreen")
plt.title("Purchase Amount Violin Plot")
plt.show()
plt.figure(figsize=(6,4))
sns.boxplot(y=df["PurchaseAmount"], color="orange")
plt.title("Purchase Amount Box Plot")
plt.show()
plt.figure(figsize=(6,4))
sns.kdeplot(df["PurchaseAmount"], fill=True, color="red")
plt.title("Purchase Amount Density Plot")
plt.show()
plt.figure(figsize=(7,4))
sns.barplot(x="Category", y="PurchaseAmount", data=df,
estimator=sum)
plt.title("Total Purchase Amount by Category")
```

```
plt.xlabel("Category")
plt.ylabel("Total Purchase Amount")
plt.show()
```

Output:-

```
  Customer  PurchaseAmount  OrderFrequency     Category
0       C1            1200               5  Electronics
1       C2            2500               8      Fashion
2       C3            1800               6  Electronics
3       C4            4200              12         Home
4       C5            3100              10      Fashion
5       C6            1500               4         Home
6       C7            5000              15  Electronics
7       C8            2700               7      Fashion
```



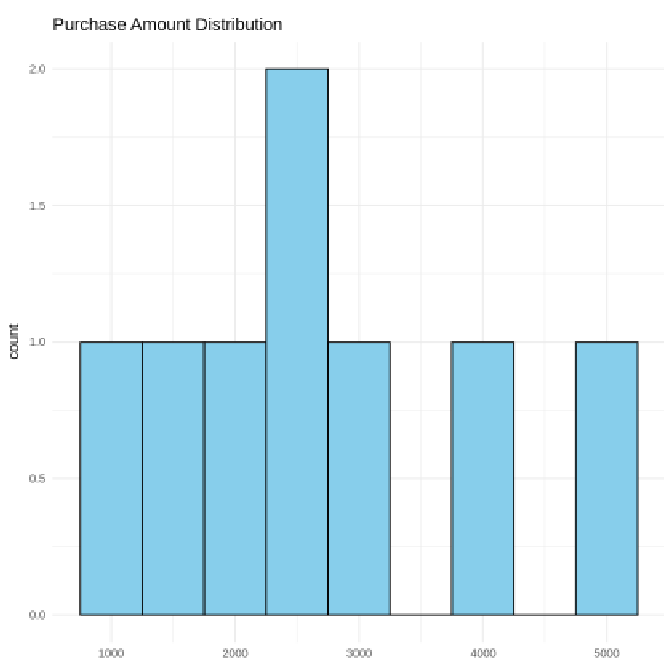
R-code :-

```r
library(ggplot2)

library(dplyr)

customer <- data.frame(

  Customer = c("C1","C2","C3","C4","C5","C6","C7","C8"),

  PurchaseAmount =
c(1200,2500,1800,4200,3100,1500,5000,2700),

  OrderFrequency = c(5,8,6,12,10,4,15,7),

  Category = c("Electronics","Fashion","Electronics","Home",

            "Fashion","Home","Electronics","Fashion")

)

print(customer)

ggplot(customer, aes(x = PurchaseAmount)) +

  geom_histogram(binwidth = 500, fill = "skyblue", color =
"black") +

  ggtitle("Purchase Amount Distribution") +

  theme_minimal()

ggplot(customer, aes(x = "", y = PurchaseAmount)) +

  geom_violin(fill = "lightgreen") +

  ggtitle("Purchase Amount Violin Plot") +
```

```
  xlab("") +
  theme_minimal()
ggplot(customer, aes(x = "", y = PurchaseAmount)) +
  geom_boxplot(fill = "orange") +
  ggtitle("Purchase Amount Box Plot") +
  xlab("") +
  theme_minimal()
ggplot(customer, aes(x = PurchaseAmount)) +
  geom_density(fill = "pink", alpha = 0.5) +
  ggtitle("Purchase Amount Density Plot") +
  theme_minimal()
ggplot(customer, aes(x = Category, y = PurchaseAmount, fill
= Category)) +
  stat_summary(fun = sum, geom = "bar") +
  ggtitle("Total Purchase Amount by Category") +
  theme_minimal()
```

Output:-

	Customer	PurchaseAmount	OrderFrequency	Category
1	C1	1200	5	Electronics
2	C2	2500	8	Fashion
3	C3	1800	6	Electronics
4	C4	4200	12	Home
5	C5	3100	10	Fashion
6	C6	1500	4	Home
7	C7	5000	15	Electronics
8	C8	2700	7	Fashion



3.

Python code :-

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns


data = {

   "Department": ["CSE", "CSE", "AI&DS", "AI&DS", "ECE",
"ECE", "MECH", "MECH"],

   "Student": ["S1", "S2", "S3", "S4", "S5", "S6", "S7", "S8"],

   "Marks": [92, 85, 78, 95, 88, 72, 80, 90]

}


df = pd.DataFrame(data)

print(df)

plt.figure(figsize=(7,5))

sns.barplot(x="Department", y="Marks", data=df,
estimator="mean")

plt.title("Average Marks by Department")

plt.xlabel("Department")
```

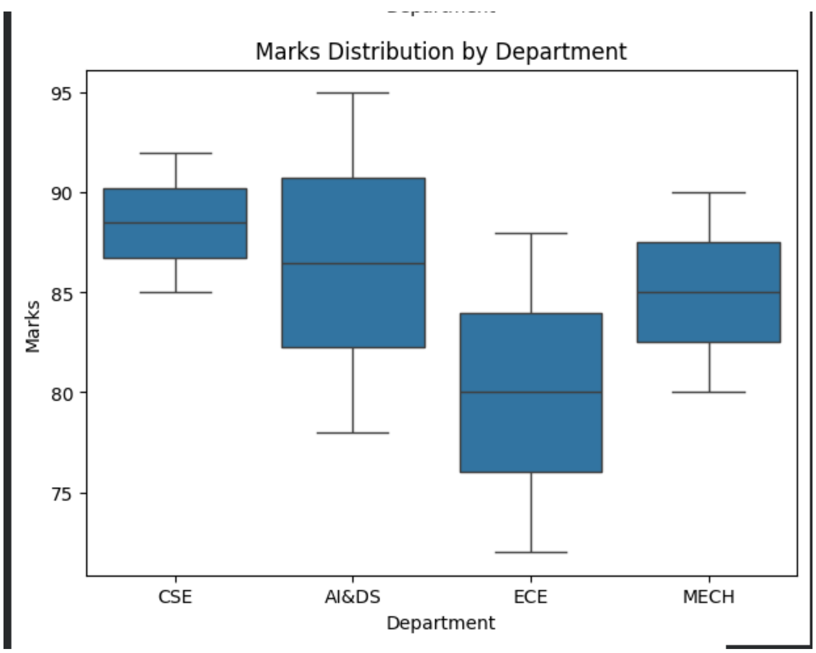
```
plt.ylabel("Average Marks")

plt.show()

plt.figure(figsize=(7,5))

sns.boxplot(x="Department", y="Marks", data=df)

plt.title("Marks Distribution by Department")

plt.show()

plt.figure(figsize=(6,4))

sns.histplot(df["Marks"], bins=5, kde=True, color="skyblue")

plt.title("Distribution of Student Marks")

plt.xlabel("Marks")

plt.show()

plt.figure(figsize=(7,5))

sns.scatterplot(x="Department", y="Marks",
hue="Department", s=100, data=df)

plt.title("Student Marks by Department")

plt.show()

pivot = df.pivot_table(values="Marks", index="Department",
aggfunc="mean")

plt.figure(figsize=(5,4))

sns.heatmap(pivot, annot=True, cmap="YlGnBu")
```

plt.title("Average Marks Heatmap")

plt.show()

Output:-

	Department	Student	Marks
0	CSE	S1	92
1	CSE	S2	85
2	AI&DS	S3	78
3	AI&DS	S4	95
4	ECE	S5	88
5	ECE	S6	72



R code :-

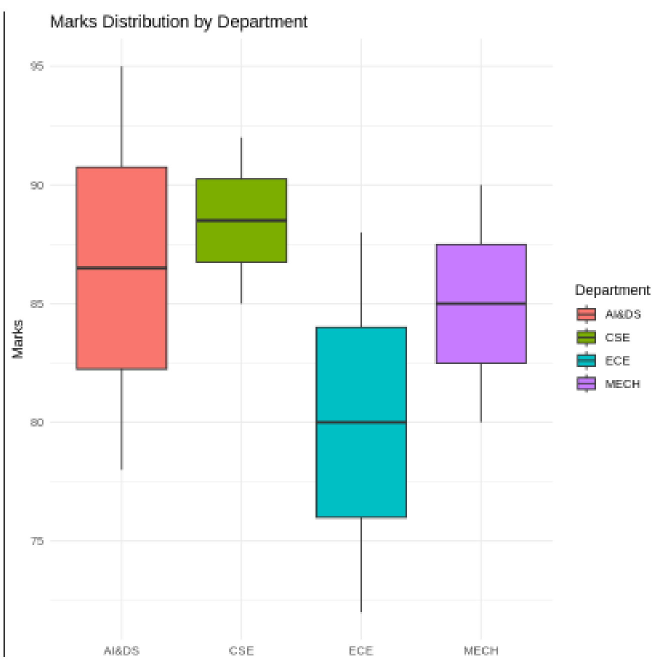
```
library(ggplot2)
library(dplyr)
students <- data.frame(
  Department =
c("CSE","CSE","AI&DS","AI&DS","ECE","ECE","MECH","MECH"),
  Student = c("S1","S2","S3","S4","S5","S6","S7","S8"),
  Marks = c(92,85,78,95,88,72,80,90)
)
print(students)
ggplot(students, aes(x = Department, y = Marks, fill =
Department)) +
  stat_summary(fun = mean, geom = "bar") +
  ggtitle("Average Marks by Department") +
  theme_minimal()
ggplot(students, aes(x = Department, y = Marks, fill =
Department)) +
  geom_boxplot() +
  ggtitle("Marks Distribution by Department") +
```

```
  theme_minimal()
ggplot(students, aes(x = Marks)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
  ggtitle("Distribution of Student Marks") +
  theme_minimal()
ggplot(students, aes(x = Department, y = Marks, color = Department)) +
  geom_point(size = 4) +
  ggtitle("Student Marks by Department") +
  theme_minimal()
avg_marks <- students %>%
  group_by(Department) %>%
  summarise(Average_Marks = mean(Marks))

ggplot(avg_marks, aes(x = "Average", y = Department, fill = Average_Marks)) +
  geom_tile() +
  geom_text(aes(label = round(Average_Marks, 1)), color = "white", size = 5) +
```

```
ggtitle("Average Marks Heatmap") +
theme_minimal()
```

Output:-

```
  Department Student Marks
1        CSE      S1    92
2        CSE      S2    85
3      AI&DS      S3    78
4      AI&DS      S4    95
5        ECE      S5    88
6        ECE      S6    72
7       MECH      S7    80
```



4.

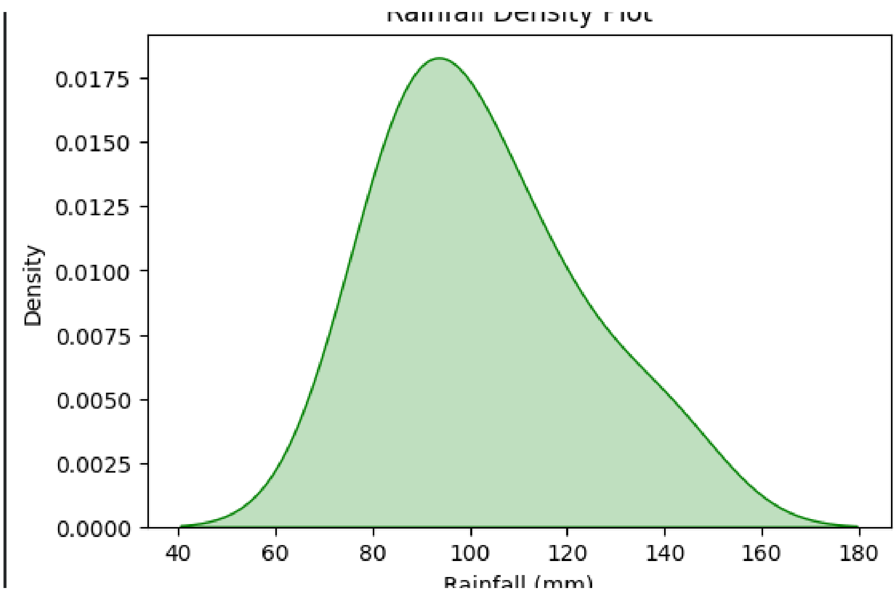
Python code :-

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

data = {

   "City": ["Chennai", "Coimbatore", "Madurai", "Salem",

            "Trichy", "Chennai", "Coimbatore", "Madurai"],

   "Rainfall": [120, 95, 80, 110, 90, 140, 100, 85],

   "Temperature": [34, 30, 36, 33, 35, 32, 29, 37],

   "Humidity": [78, 70, 65, 72, 68, 82, 74, 63]

}

df = pd.DataFrame(data)

print(df)

plt.figure(figsize=(6,4))

sns.histplot(df["Rainfall"], bins=5, kde=True,
color="skyblue")

plt.title("Rainfall Distribution")

plt.xlabel("Rainfall (mm)")
```

```
plt.show()

plt.figure(figsize=(6,4))

sns.kdeplot(df["Rainfall"], fill=True, color="green")

plt.title("Rainfall Density Plot")

plt.xlabel("Rainfall (mm)")

plt.show()

plt.figure(figsize=(6,4))

sns.boxplot(y=df["Rainfall"], color="orange")

plt.title("Rainfall Box Plot")

plt.ylabel("Rainfall (mm)")

plt.show()

plt.figure(figsize=(7,5))

sns.barplot(x="City", y="Rainfall", data=df,
estimator="mean")

plt.title("Average Rainfall by City")

plt.xlabel("City")

plt.ylabel("Average Rainfall (mm)")

plt.show()
```

Output:-

	City	Rainfall	Temperature	Humidity
0	Chennai	120	34	78
1	Coimbatore	95	30	70
2	Madurai	80	36	65
3	Salem	110	33	72
4	Trichy	90	35	68
5	Chennai	140	32	82
6	Coimbatore	100	29	74



R-code :-

```
# Load Libraries
library(ggplot2)
library(dplyr)

# Create Dataset
climate <- data.frame(
  City = c("Chennai", "Coimbatore", "Madurai", "Salem",
           "Trichy", "Chennai", "Coimbatore", "Madurai"),
  Rainfall = c(120, 95, 80, 110, 90, 140, 100, 85),
  Temperature = c(34, 30, 36, 33, 35, 32, 29, 37),
  Humidity = c(78, 70, 65, 72, 68, 82, 74, 63)
)

# Display Dataset
print(climate)

# -------------------------
```

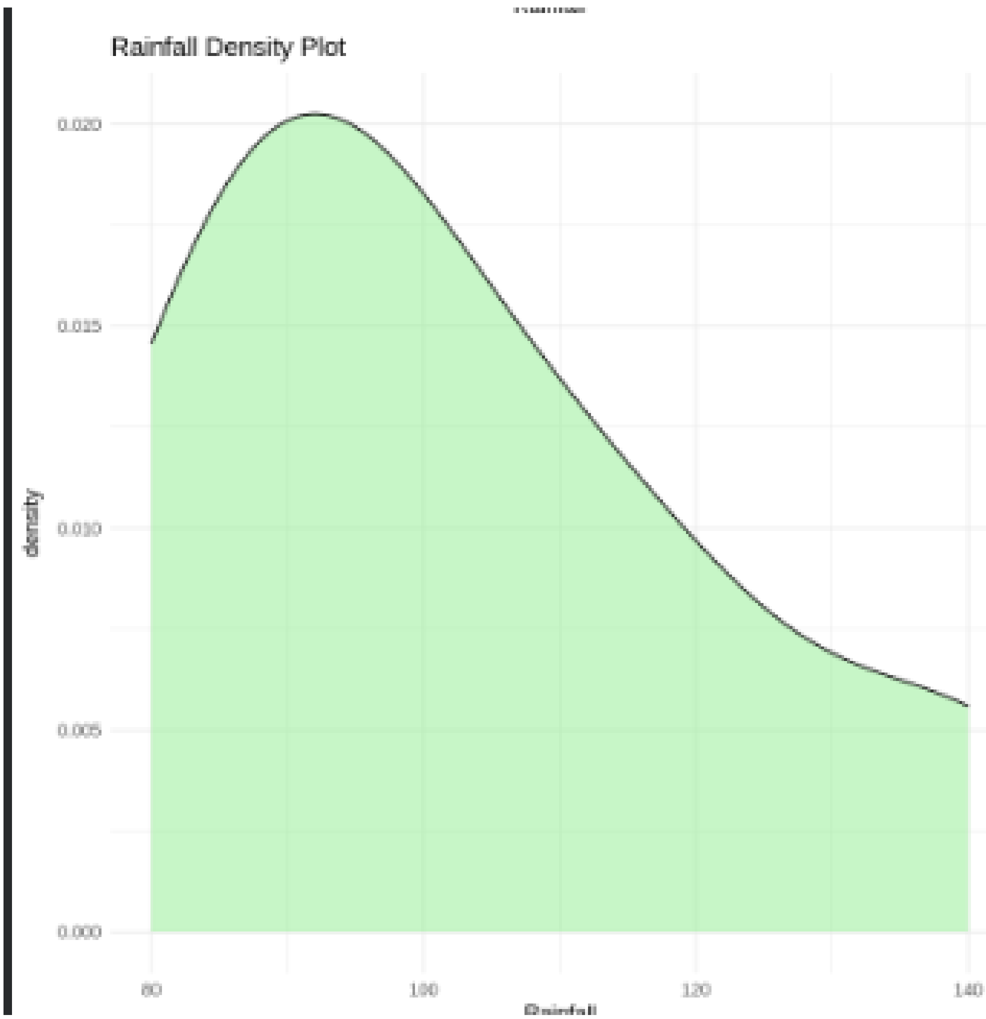
```
# 1. Histogram
# -------------------------
ggplot(climate, aes(x = Rainfall)) +
  geom_histogram(binwidth = 10, fill = "skyblue", color = "black") +
  ggtitle("Rainfall Distribution") +
  theme_minimal()


# -------------------------
# 2. Density Plot
# -------------------------
ggplot(climate, aes(x = Rainfall)) +
  geom_density(fill = "lightgreen", alpha = 0.5) +
  ggtitle("Rainfall Density Plot") +
  theme_minimal()


# -------------------------
# 3. Box Plot
# -------------------------
```

```
ggplot(climate, aes(y = Rainfall)) +
  geom_boxplot(fill = "orange") +
  ggtitle("Rainfall Box Plot") +
  theme_minimal()

# --------------------------
# 4. Bar Chart
# --------------------------
ggplot(climate, aes(x = City, y = Rainfall, fill = City)) +
  stat_summary(fun = mean, geom = "bar") +
  ggtitle("Average Rainfall by City") +
  theme_minimal()
```

Output:-

```
        City Rainfall Temperature Humidity
1    Chennai      120          34       78
2 Coimbatore       95          30       70
3    Madurai       80          36       65
4      Salem      110          33       72
```

5	Trichy	90	35	68
6	Chennai	140	32	82
7	Coimbatore	100	29	74
8	Madurai	85	37	63



5.

Python code :-

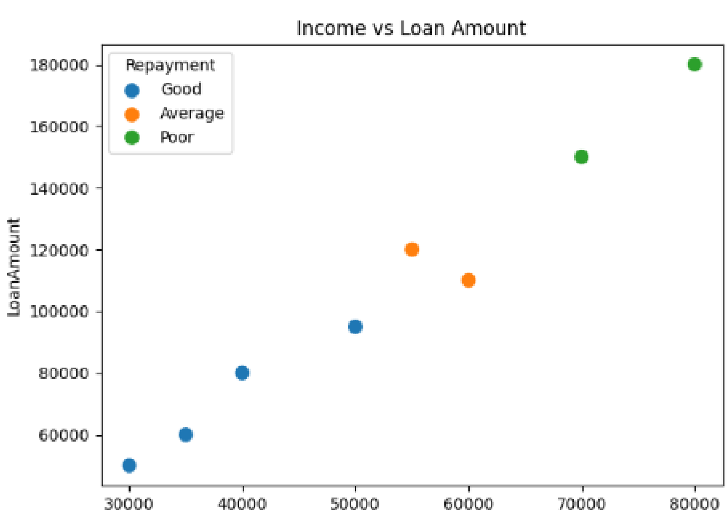
```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns


# Create Dataset

data = {

   "Customer": ["C1", "C2", "C3", "C4", "C5", "C6", "C7", "C8"],

   "LoanAmount": [50000, 120000, 80000, 150000, 95000,
60000, 180000, 110000],

   "Income": [30000, 55000, 40000, 70000, 50000, 35000,
80000, 60000],

   "CreditScore": [720, 680, 750, 650, 710, 730, 620, 690],

   "Repayment": ["Good", "Average", "Good", "Poor", "Good",
"Good", "Poor", "Average"]

}
```

```
df = pd.DataFrame(data)

print(df)

# -------------------------
# 1. Histogram
# -------------------------
plt.figure(figsize=(6,4))
sns.histplot(df["LoanAmount"], bins=5, kde=True,
color="skyblue")
plt.title("Loan Amount Distribution")
plt.xlabel("Loan Amount")
plt.show()

# -------------------------
# 2. Box Plot
# -------------------------
plt.figure(figsize=(6,4))
sns.boxplot(y=df["LoanAmount"], color="orange")
```

```
plt.title("Loan Amount Box Plot")

plt.ylabel("Loan Amount")

plt.show()


# -------------------------

# 3. Scatter Plot

# -------------------------

plt.figure(figsize=(7,5))

sns.scatterplot(x="Income", y="LoanAmount",
hue="Repayment", s=100, data=df)

plt.title("Income vs Loan Amount")

plt.show()


# -------------------------

# 4. Bar Chart

# -------------------------

plt.figure(figsize=(6,4))

sns.barplot(x="Repayment", y="LoanAmount", data=df,
estimator="mean")

plt.title("Average Loan Amount by Repayment Status")
```

```
plt.show()


# -------------------------
# 5. Credit Score Distribution
# -------------------------
plt.figure(figsize=(6,4))
sns.histplot(df["CreditScore"], bins=5, color="green", kde=True)
plt.title("Credit Score Distribution")
plt.xlabel("Credit Score")
plt.show()
```

Output :-



R-code:-

```
# Load Libraries
library(ggplot2)
library(dplyr)

# Create Dataset
loan <- data.frame(
  Customer = c("C1","C2","C3","C4","C5","C6","C7","C8"),
  LoanAmount =
c(50000,120000,80000,150000,95000,60000,180000,11000
0),
  Income =
c(30000,55000,40000,70000,50000,35000,80000,60000),
  CreditScore = c(720,680,750,650,710,730,620,690),
  Repayment =
c("Good","Average","Good","Poor","Good","Good","Poor","Avera
ge")
)

# Display Dataset
```

```r
print(loan)

# -------------------------
# 1. Histogram
# -------------------------
ggplot(loan, aes(x = LoanAmount)) +
  geom_histogram(binwidth = 25000, fill = "skyblue", color = "black") +
  ggtitle("Loan Amount Distribution") +
  theme_minimal()

# -------------------------
# 2. Box Plot
# -------------------------
ggplot(loan, aes(y = LoanAmount)) +
  geom_boxplot(fill = "orange") +
  ggtitle("Loan Amount Box Plot") +
  theme_minimal()
```

```r
# -------------------------
# 3. Scatter Plot
# -------------------------
ggplot(loan, aes(x = Income, y = LoanAmount, color = Repayment)) +
  geom_point(size = 4) +
  ggtitle("Income vs Loan Amount") +
  theme_minimal()


# -------------------------
# 4. Bar Chart
# -------------------------
ggplot(loan, aes(x = Repayment, y = LoanAmount, fill = Repayment)) +
  stat_summary(fun = mean, geom = "bar") +
  ggtitle("Average Loan Amount by Repayment Status") +
  theme_minimal()


# -------------------------
# 5. Credit Score Distribution
```

```
ggplot(loan, aes(x = CreditScore)) +
  geom_histogram(binwidth = 20, fill = "lightgreen", color = 
"black") +
  ggtitle("Credit Score Distribution") +
  theme_minimal()
```

Output:-